

Underwater Vision-Based Surveillance Using Machine Learning and Artificial Intelligence

Day 27 — Final Mentor Review

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Project Overview

Objective

- Analyze all three supplied underwater datasets
- Perform image preprocessing and enhancement
- Apply YOLOv8n for object detection
- Compare original and enhanced images
- Evaluate detection performance, runtime and model size
- Analyze prediction errors and limitations

Research Question

How does underwater image enhancement affect the performance of deep-learning-based object detection across the three supplied underwater datasets?

Datasets & Final Scope

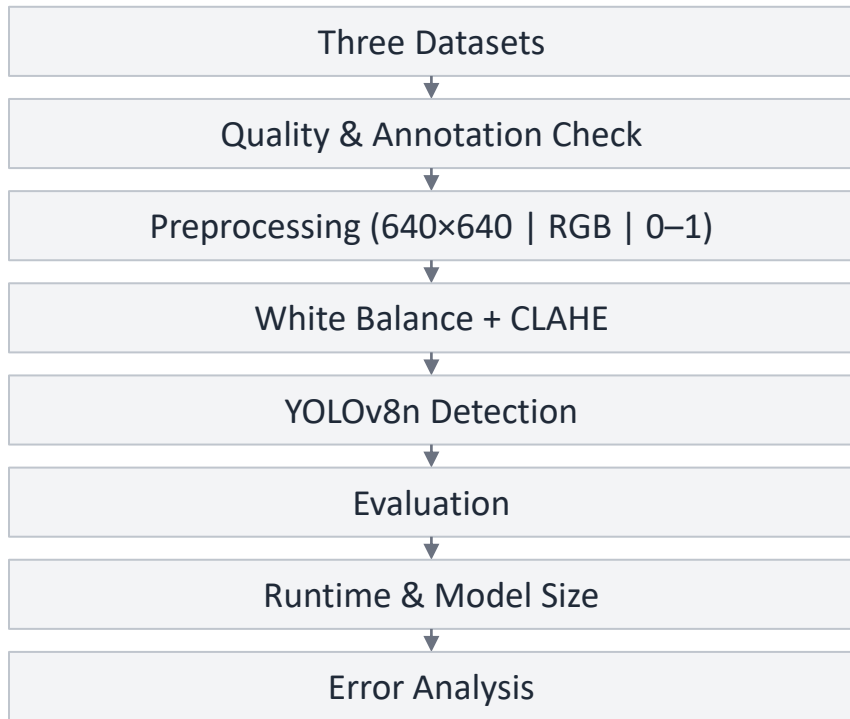
Dataset	Classes	Final Configuration
DIATAquarium	11	Original + YOLOv8n
Aquatic Plant	1	WB+CLAHE + YOLOv8n
Well	4	WB+CLAHE + YOLOv8n

Scope

- All three datasets evaluated separately
- YOLOv8n used for object detection
- White Balance + CLAHE implemented for enhancement
- Tracking excluded because video data was not available

Final Methodology

Pipeline



Evaluation Metrics

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95
- Inference Time
- Model Size

Final Model Configurations

DIATAquarium

Original → YOLOv8n

- Batch size: 16
- Epochs: 30
- Image size: 640

Aquatic Plant — H03

Original → WB+CLAHE → YOLOv8n

- Learning rate: 0.0005
- Batch size: 32
- Epochs: 30
- Optimizer: AdamW

Well — H04

Original → WB+CLAHE → YOLOv8n

- Learning rate: 0.001
- Batch size: 16
- Epochs: 30
- Optimizer: AdamW

Final Validation Results

Dataset	Precision	Recall	mAP@0.5	mAP@0.5:0.95
DIAT	90.02%	92.09%	92.73%	65.92%
Aquatic Plant	95.4%	96.1%	98.5%	71.5%
Well	35.7%	40.1%	36.8%	16.8%

Note: DIAT results use the available validation set because the supplied local copy did not contain a separate test set.

Final Test Results & Performance

Test Results

Dataset	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Aquatic Plant	97.99%	90.26%	96.75%	71.12%
Well	43.00%	37.46%	39.34%	16.15%

Runtime

Dataset	Time/Image	FPS	Model Size
DIAT	22.14 ms	45.16	5.95 MB
Aquatic Plant	9.80 ms	102.02	5.96 MB
Well	9.98 ms	100.21	5.96 MB

Software Artefacts & Repository

Final Project Package

```
Day26_Final_Project/  
├── preprocessing/  
├── enhancement/  
├── detection/  
├── evaluation/  
├── models/  
├── results/  
├── figures/  
├── docs/  
├── README.md  
└── requirements.txt
```

Repository

<https://github.com/Nitesh71332/Underwater-vision-surveillance>

Included

- Source code (preprocessing, enhancement, YOLO inference)
- Evaluation results
- Selected figures and prediction outputs
- Error-analysis examples
- Project documentation
- README and requirements file

Error Analysis & Limitations

DIAT

- Fish/Fishes confusion
- Possible missed Camera detection
- Low-confidence predictions

Aquatic Plant

- Missed plant detections
- Low-confidence predictions
- Haze and unclear boundaries

Well

- Small and edge-located objects
- Difficult stone detections
- Class imbalance
- Possible false positives

Limitations

- DIAT local test set unavailable
- Source overlap identified
- Well has strong class imbalance
- Error analysis covered representative difficult cases
- Water-Net/FUnIE-GAN reviewed, not implemented
- Tracking excluded (no video data)

Mentor Approval & Technical Sign-Off

Mentor Review Outcome

Review Area	Status
Methodology	✔ Approved
Dataset Handling	✔ Approved
Preprocessing	✔ Approved
Enhancement Method	✔ Approved
Model Configuration	✔ Approved
Results	✔ Approved
Error Analysis	✔ Approved
Limitations	✔ Approved
PPT	✔ Approved
Repository	✔ Approved

Mentor Corrections

No major corrections required.

Action Taken

Final methodology, results, software artefacts, limitations, PPT and repository structure were reviewed and accepted.

Technical Sign-Off

Mentor Review Status: ✔ Approved

Technical Results: ✔ Frozen

Final Status: Day 27 Completed